



Kasuwa Credit OS

CREDITCOIN CTC 2026 FALL · RWA TRACK

Verified economic activity becomes reusable, programmable credit.

The credit state layer for Creditcoin — not just another lending app. Kasuwa turns independently attested, real-world merchant activity into on-chain credit capacity that any protocol can build on.

THE PROBLEM

Real economic activity is invisible to on-chain credit.

Today

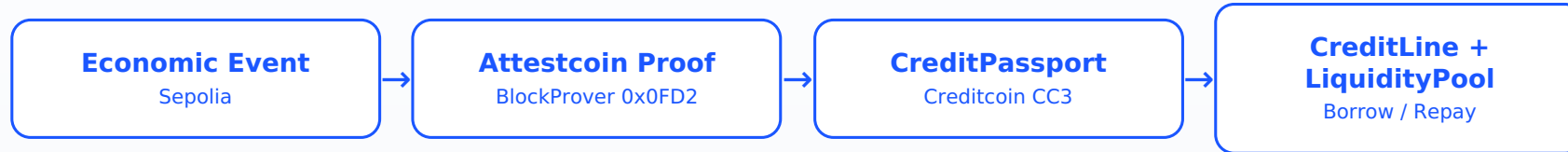
Merchants in emerging markets (starting with Nigeria) generate real, verifiable transaction history every day — sales, repayments, verified fulfillment. On-chain lenders can't see any of it. Most DeFi credit still requires crypto collateral, shutting out the population with the most real economic activity to underwrite against.

With Kasuwa

Off-chain economic events are independently attested cross-chain via the Attestcoin Protocol, recorded on-chain, and converted into a reusable credit passport — capacity that any Creditcoin protocol can lend against, without a bridge or a custodian in the middle.

"Not just another isolated lending app — a credit state layer other protocols build on."

From a real-world event to on-chain credit capacity



Cross-chain verification

EconomicEvents.sol emits events on Ethereum Sepolia. A relay watches and submits proofs through Attestcoin's BlockProver precompile, verified on Creditcoin CC3 by AttestcoinVerifier.sol before anything is recorded.

Reusable credit capacity

CreditPassport.sol records verified history and recomputes capacity deterministically. CreditLine.sol and LiquidityPool.sol let merchants borrow and repay against a shared pool any integrator can plug into.

We audited our own threat model — and fixed what we found.

The finding

Internal review of our own threat model surfaced a real reentrancy exposure in `CreditLine.borrow()`: an external call executed before state was updated.

The fix

- Checks-effects-interactions ordering: state is recorded before external calls
- A dependency-free ReentrancyGuard on borrow / repay / withdraw / fundCreditLine
- Dedicated attacker-contract tests proving the exploit is blocked

45 / 45

tests passing

1

critical reentrancy bug found & fixed

2

chains — Sepolia + Creditcoin CC3

Not a local demo — a live, verifiable deployment.

A GitHub Actions pipeline deploys the source-chain contract on Ethereum Sepolia and the full Kasuwa stack on Creditcoin CC3 testnet, with a relayer that watches for events and submits real cross-chain Attestcoin proofs.

Component	Chain	Purpose
EconomicEvents.sol	Ethereum Sepolia	Emits the source-chain economic event
AttestcoinVerifier.sol	Creditcoin CC3	Validates the cross-chain proof
CreditPassport.sol	Creditcoin CC3	Records verified history, recomputes capacity
CreditLine.sol / LiquidityPool.sol	Creditcoin CC3	Borrow / repay against shared liquidity
Relayer service	Off-chain	Watches source-chain events, submits proofs

A credit state layer, not another lending app.

Composable by design

Credit capacity lives in CreditPassport as reusable state. Any protocol on Creditcoin can read and build against it — not locked inside one app's UI.

Collateral-free for real activity

Merchants borrow against verified economic history, not locked crypto — opening credit to the population DeFi has structurally excluded.

Security-first, honestly documented

We disclose the reentrancy bug we found in our own threat model, the fix, and the tests that prove it — not just a clean surface.

Evidence over claims

Real Sepolia + Creditcoin CC3 testnet deployment via CI, with on-chain transactions judges can verify themselves.

Try it, read it, verify it.

Live demo: <https://kasuwa-credit-os.vercel.app/>

GitHub repo: <https://github.com/Xzavior34/kasuwa-credit-os>

Tracks: RWA (primary), DeFi (secondary)

Tags: Creditcoin, Attestcoin Protocol, RWA, DeFi, Solidity, EVM

Team: Sara Ventures / Sara Foundation Africa

Built for Creditcoin CTC 2026 Fall — BUIDL For The Real World.